Estimating Implicit Regularization in Deep Learning

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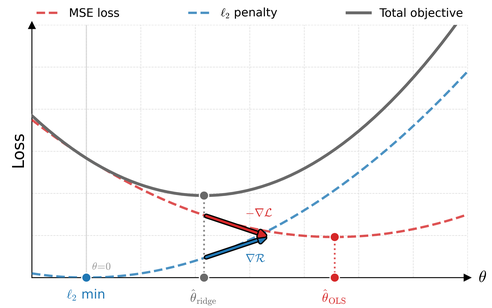
Reverse-engineer the bias

Why do overparameterized networks generalize?

Deep networks can memorize training data yet still generalize to new examples [1].

Explicit regularization adds penalties that favor simpler solutions and can improve generalization. **Implicit regularization** arises from training and architectural choices, biasing which solutions are learned [2].

Can we estimate an explicit penalty that mimics an implicit bias?



Fix the weights; estimate the penalty

At a hypothetical minimum of $\mathcal{L}+\mathcal{R}$, the gradients counteract one another:

$$-\nabla_\theta \mathcal{L}(\hat{\theta}, D) = \nabla_\theta \mathcal{R}(\hat{\theta}, D; \Lambda).$$

Fit the regularizer parameters with the weights held fixed:

$$\hat{\Lambda} = \arg\min_{\Lambda} \frac{1}{p} \left\| \nabla_\theta \mathcal{L} + \nabla_\theta \mathcal{R}_\Lambda \right\|_2^2.$$

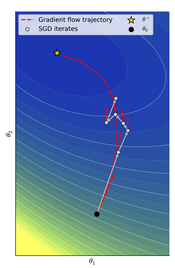
Follow the trajectory, not just the endpoint

SGD iterates deviate from the full-batch gradient-flow path (right). For a gradient-step reference, match the update residual:

$$-\frac{\Delta\theta_t}{\eta} - \nabla_\theta \mathcal{L}(\theta_t) \;\approx\; \nabla_\theta \mathcal{R}(\theta_t; \Lambda),$$

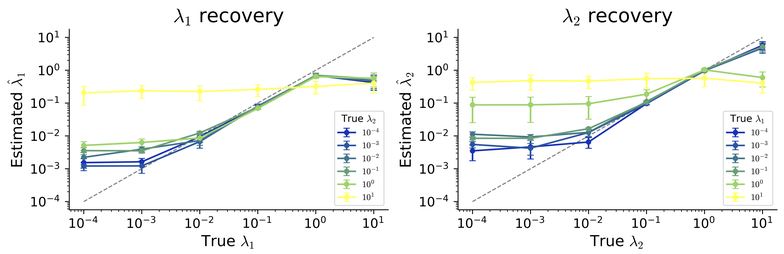
where $\Delta\theta_t = \theta_{t+1} - \theta_t$.

Stack steps to estimate a shared regularizer, or fit time-varying parameters. Setting the update to zero recovers the endpoint perspective.



Recover known effects

Explicit ℓ_1 and ℓ_2 penalties



Deep ReLU networks; 6×6 penalty grid, 10 datasets per cell. Mean ± SE; dashed line = exact recovery.

Recovery is accurate at intermediate strengths. Weak signals and nearly collinear gradients limit identification at the extremes.

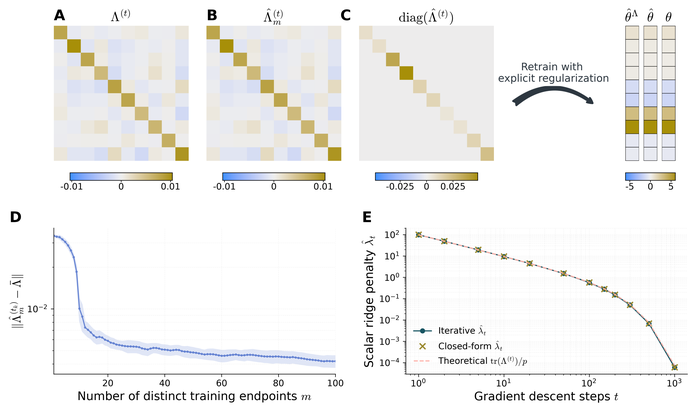
Early stopping induces a quadratic penalty

The early-stopped OLS iterate solves a regularized problem [3]:

$$\hat{\theta}_t = \arg\min_{\theta} \left\{ \frac{1}{n}\|y - X\theta\|_2^2 + \mathcal{R}_t(\theta) \right\},$$

$$\mathcal{R}_t(\theta) = \theta^\top \Lambda^{(t)} \theta, \qquad \Lambda^{(t)} = VS\left[(I-\eta S)^{-t} - I\right]^{-1} V^\top.$$

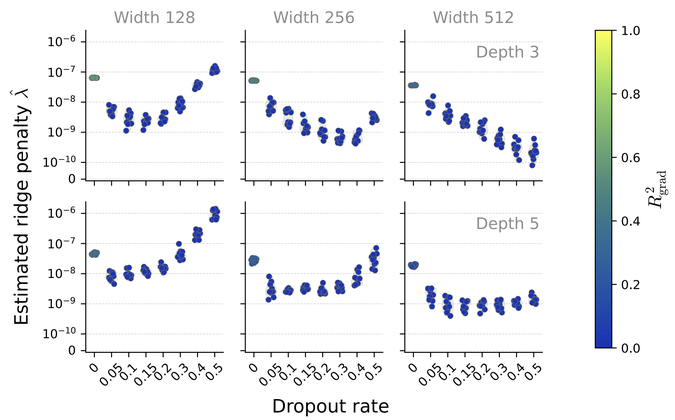
Here $X^\top X/n = VSV^\top$, $\theta_0 = 0$, and $0 < \eta s_i < 1$. The step convention is $\theta_{t+1} = \theta_t + \eta X^\top(y - X\theta_t)/n$.



A–B: recover the theoretical matrix. **C:** a diagonal fit reproduces weights. **D–E:** more endpoints improve estimation; the penalty decays over training.

Implicit bias in networks

Dropout has data-dependent regularization effects [4]. On MNIST, vary depth, width, and dropout rate; fit $\mathcal{R}(\theta, \lambda) = \lambda\|\theta\|_2^2$.

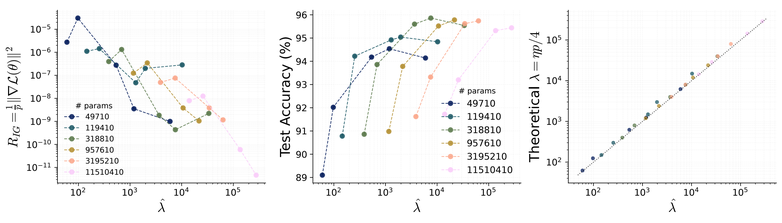


Each point is one of 10 seeds. Color shows gradient fit: lighter = more explained. Zero-dropout controls are included.

Narrow networks show an upturn at high dropout; wider networks have smaller estimates. Scalar ridge summarizes part of the effect.

Recover regularization along trajectories

Barrett & Dherin predict $\mathcal{R} = \frac{\lambda}{p}\|\nabla\mathcal{L}\|^2$ with $\lambda = \eta p/4$ [5]. Estimates from deviations of discrete GD from continuous gradient flow track theory (right).



Our contributions

- **Estimate implicit bias** by matching loss gradients to a proposed regularizer.
- **Characterize success:** recover known regularization or reproduce the original model's weights.
- **Bridge theory and practice:** validate analytical results and diagnose bias in complex networks.

References

[1] Chiyuan Zhang, Samy Bengio, Moritz Hardt, Benjamin Recht, and Oriol Vinyals. Understanding deep learning (still) requires rethinking generalization. *Communications of the ACM*, 2021.

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[3] Alnur Ali, J. Zico Kolter, and Ryan J. Tibshirani. A continuous-time view of early stopping for least squares regression. In *AISTATS*, 2019.

[4] Colin Wei, Sham Kakade, and Tengyu Ma. The implicit and explicit regularization effects of dropout. In *ICML*, 2020.

[5] David G. T. Barrett and Benoit Dherin. Implicit gradient regularization, 2022. arXiv:2009.11162.

Paper: arXiv:2605.05436 **Blog:** jrudoler.com/estimating-implicit-regularization/ **Code:** github.com/jrudoler/estimating-implicit-regularization

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